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Math 102 Final Exam Term 231

Original exam date: 20 December 2023. Practice version: questions and choices shuffled.
Net time allowed: 180 minutes

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1. The upper sum for the region bounded by the graph of $f(x) = x^2$ and the x-axis between $x = 0$ and $x = 2$ in terms of n (the number of subintervals) is

(a) $\frac{8}{3} - \frac{4}{n} + \frac{4}{3n^2}$

(b) $\frac{8}{3} + \frac{4}{3n^2}$

(c) $\frac{5}{3} + \frac{2}{n} + \frac{2}{3n^2}$

(d) $\frac{8}{3} + \frac{4}{n} + \frac{4}{3n^2}$

(e) $\frac{5}{3} - \frac{2}{n} + \frac{2}{3n^2}$

2. Consider the following series I. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^{4/3}}$ II. $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+4}}$. Then

(a) I converges absolutely but II diverges

(b) I and II converge conditionally

(c) I diverges but II converges absolutely

(d) I and II converge absolutely

(e) I converges absolutely but II converges conditionally

3. The area of the region bounded by the graphs of the equations $y = xe^{-x^2/4}$, $y = 0$, $x = 0$ and $x = \sqrt{6}$ is

(a) $1 - e^{-3/2}$

(b) $2 - 2e^{-3/2}$

(c) $e^{-3/2}$

(d) $2 + 2e^{-3/2}$

(e) $-2 - 2e^{-3/2}$

4. If $P_3(x) = 2 + b(x-1) + c(x-1)^2 + d(x-1)^3$ is the 3rd Taylor polynomial for the function $f(x) = \dfrac{2}{x}$, centered at $x = 1$, then $b + c + d =$

(a) 0

(b) 4

(c) 2

(d) -2

(e) -4

5. Which of the following series is convergent? I. $\sum_{n=1}^{\infty} \frac{(-1)^n n}{\ln(n+1)}$ II. $\sum_{n=1}^{\infty} \sin \frac{(2n-1)\pi}{2}$ III. $\sum_{n=1}^{\infty} \frac{1}{n} \cos n\pi$

(a) I and II only

(b) I, II and III

(c) III only

(d) I and III only

(e) I only

6. Which of the following series is convergent? I. $\sum_{n=0}^{\infty} \frac{1}{n!}$ II. $\sum_{n=1}^{\infty} \sin \frac{1}{n}$ III. $\sum_{n=0}^{\infty} \frac{n!}{2(n!)+1}$

(a) I and II only

(b) I only

(c) II and III only

(d) I and III only

(e) I, II and III

7. $\int_0^2 \frac{1}{(x-1)^2}\, dx$

(a) is equal to 2

(b) is equal to 0

(c) diverges

(d) is equal to -2

(e) is equal to 1

8. $\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}}\,dx =$

(a) 0

(b) $\dfrac{\pi}{4}$

(c) $\dfrac{\pi}{2}$

(d) $-\dfrac{\pi}{2}$

(e) $-\dfrac{\pi}{4}$

9. Using the integral test, and for positive values of p, the series $\sum_{n=1}^{\infty} \frac{n}{(1+n^2)^p}$ converges if

(a) $p > 1$

(b) $p \geq 1$

(c) $0 < p \leq 1$

(d) $p > 0$

(e) $0 < p < 1$

10. $\int \frac{dx}{1+\sqrt{2x}} =$

(a) $\sqrt{2x} - \ln\left(1+\sqrt{2x}\right) + c$

(b) $\ln\left(1+\sqrt{2x}\right) + c$

(c) $\sqrt{2x} + c$

(d) $\sqrt{2x} + \ln\left(1+\sqrt{2x}\right) + c$

(e) $\sqrt{x} - \ln\left(1+\sqrt{x}\right) + c$

11. The power series representation for the function $f(x) = \dfrac{1+x}{(1-x)^2}$ is

(a) $\sum_{n=0}^{\infty}(2n+1)x^n,\ |x| < 1$

(b) $\sum_{n=0}^{\infty}(2n-1)x^n,\ |x| < 1$

(c) $\sum_{n=0}^{\infty}(n-1)x^n,\ |x| < 1$

(d) $\sum_{n=0}^{\infty}(n+1)x^n,\ |x| < 1$

(e) $\sum_{n=0}^{\infty}(2n+1)x^{n+1},\ |x| < 1$

12. $\sum_{n=0}^{\infty}(0.3)^n =$

(a) $\frac{10}{7}$

(b) $\frac{9}{7}$

(c) $\frac{11}{7}$

(d) $\frac{8}{7}$

(e) $\frac{12}{7}$

13. Using the Root test for the series $\sum_{n=1}^{\infty} a_n$ where $a_n = \frac{n}{3^n}$, we have $\lim_{n\to\infty} \sqrt[n]{|a_n|} =$

(a) $\frac{2}{3}$

(b) $\frac{1}{2}$

(c) $\frac{1}{3}$

(d) 1

(e) $\frac{4}{3}$

14. $\int \frac{xe^{2x}}{(2x+1)^2}\,dx =$

(a) $\frac{e^{2x}}{2(2x+1)} + c$

(b) $\frac{e^{2x}}{4(x+1)} + c$

(c) $\frac{e^{x}}{2(2x+1)} + c$

(d) $\frac{e^{x}}{4(2x+1)} + c$

(e) $\frac{e^{2x}}{4(2x+1)} + c$

15. Given that the sequence $\sqrt{2},\ \sqrt{2+\sqrt{2}},\ \sqrt{2+\sqrt{2+\sqrt{2}}}, \ldots$ is a convergent sequence, its limit is equal to

(a) $\sqrt{3}$

(b) 2

(c) $2+\sqrt{2}$

(d) 3

(e) $\sqrt{2}$

16. $\int \frac{2\,dx}{\sqrt{4x-x^2}} =$

(a) $2\arcsin\left(\dfrac{x-2}{2}\right)+c$

(b) $4\arcsin(x-2)+c$

(c) $\arcsin\left(\dfrac{x-2}{2}\right)+c$

(d) $2\arcsin\left(\dfrac{x+2}{2}\right)+c$

(e) $\arcsin\left(\dfrac{x+2}{2}\right)+c$

17. The volume of the solid generated by revolving the region bounded by the graphs of the equations $y=2x-x^2$ and $y=0$ about the line $x=4$ is

(a) 6π

(b) 8π

(c) 4π

(d) 10π

(e) 12π

18. $\int_0^2 \frac{x}{1+e^{-x^2}}\,dx =$

(a) $\ln\left(\dfrac{1+e^4}{2}\right)$

(b) $\dfrac{1}{2}\ln\left(\dfrac{1+e^4}{2}\right)$

(c) $\dfrac{1}{2}\ln\left(\dfrac{1+e^{-4}}{2}\right)$

(d) $\dfrac{1}{2}\ln\left(1+e^{-4}\right)$

(e) $\dfrac{1}{2}\ln\left(1+e^4\right)$

19. $\int \cosh^2(x-1)\sinh(x-1)\,dx =$

(a) $\tanh^3(x-1)+c$

(b) $\frac{1}{3}\sinh^3(x-1)+c$

(c) $\sinh^3(x-1)+c$

(d) $\cosh^3(x-1)+c$

(e) $\frac{1}{3}\cosh^3(x-1)+c$

20. Consider the series $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^3}$. The least number of terms required to approximate the sum of the series with an error less than 0.001 is (Hint: Use Alternating Series Remainder Theorem)

(a) 14

(b) 6

(c) 8

(d) 12

(e) 10

21. The interval of convergence of the power series $\sum_{n=1}^{\infty}\frac{(-1)^{n+1}(x-4)^n}{n\,9^n}$ is

(a) $[-5,13)$

(b) $(-5,13)$

(c) $(-\infty,\infty)$

(d) $(-5,13]$

(e) $[-5,13]$

22. $\int_0^{1/2}\arctan(x^2)\,dx =$

(a) $\sum_{n=0}^{\infty}\frac{(-1)^n}{(4n-3)(2n-1)2^{4n+3}}$

(b) $\sum_{n=0}^{\infty}\frac{(-1)^n}{(4n+3)(2n+1)2^{4n+3}}$

(c) $\sum_{n=1}^{\infty}\frac{(-1)^n}{(4n+3)(2n+1)2^{4n+3}}$

(d) $\sum_{n=0}^{\infty}\frac{(-1)^n}{(4n+3)(2n+1)2^{4n-3}}$

(e) $\sum_{n=0}^{\infty}\frac{(-1)^{n+1}}{(4n+3)(2n+1)2^{4n+3}}$

23. The arc length of the graph of the function $y = \ln(\cos x)$ over the interval $\left[0, \frac{\pi}{3}\right]$ is

(a) $\ln\left(2 - \sqrt{3}\right)$

(b) $\ln\left(2 + \sqrt{3}\right)$

(c) $\ln\left(2 + \sqrt{2}\right)$

(d) $\ln\left(1 + \sqrt{3}\right)$

(e) $\ln\left(1 - \sqrt{3}\right)$

24. The sequence $\left\{\sqrt{n}\,\ln\left(1 + \frac{1}{n}\right)\right\}$

(a) converges to e

(b) converges to 0

(c) converges to 1

(d) diverges

(e) converges to 2

25. If $F(x) = \int_0^{\sin x} \sqrt{t}\,dt$, then $F'\left(\frac{\pi}{6}\right) =$

(a) $\frac{\sqrt{6}}{2}$

(b) $\frac{\sqrt{6}}{4}$

(c) $\sqrt{6}$

(d) $\frac{\sqrt{6}}{3}$

(e) $\frac{\sqrt{2}}{2}$

26. If $\frac{2x^3 - 4x - 8}{(x^2 - x)(x^2 + 4)} = \frac{2}{x} - \frac{2}{x - 1} + \frac{Cx + D}{x^2 + 4}$, then $C + D =$

(a) 10

(b) 6

(c) 8

(d) 2

(e) 4

27. $\sum_{k=1}^{\infty} \frac{1}{9k^2+3k-2} =$

(a) $\frac{1}{5}$

(b) $\frac{1}{4}$

(c) $\frac{1}{3}$

(d) $\frac{1}{7}$

(e) $\frac{1}{6}$

28. $\int \frac{x}{2}\sqrt{4+x^2}\,dx =$

(a) $\frac{1}{6}(4+x^2)^{3/2}+c$

(b) $\frac{1}{6}(2+x^2)^{3/2}+c$

(c) $\frac{1}{4}(4+x^2)^{3/2}+c$

(d) $\frac{1}{6}(4+x^2)^{1/2}+c$

(e) $\frac{1}{4}(4+x^2)^{1/2}+c$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	Right endpoints, $\Delta x = \frac{2}{n}$: $\frac{8}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{8}{3} + \frac{4}{n} + \frac{4}{3n^2}$.
2	**(e)**	I: $p = \frac{4}{3} > 1$, absolute. II: absolute series diverges ($p = \frac{1}{2}$) but AST holds: conditional.
3	**(b)**	$\int_0^{\sqrt{6}} xe^{-x^2/4}dx = \left[-2e^{-x^2/4}\right]_0^{\sqrt{6}} = 2 - 2e^{-3/2}$.
4	**(d)**	$P_3 = 2 - 2(x-1) + 2(x-1)^2 - 2(x-1)^3$, so $b + c + d = -2$.
5	**(c)**	I and II: terms do not $\to 0$; III is $\sum \frac{(-1)^n}{n}$, convergent. Only III.
6	**(b)**	I: ratio test gives $0 < 1$ (converges); II: limit comparison with $\sum \frac{1}{n}$ (diverges); III: terms $\to \frac{1}{2}$.
7	**(c)**	Infinite discontinuity at $x = 1$: $\int_0^1 \frac{dx}{(x-1)^2} = \infty$, so it diverges.
8	**(c)**	$u = e^x$: antiderivative $\arctan(e^x)$, value $\frac{\pi}{2} - 0 = \frac{\pi}{2}$.
9	**(a)**	$\int_1^\infty \frac{x\,dx}{(1+x^2)^p} = \frac{1}{2}\int u^{-p}du$ converges iff $p > 1$.
10	**(a)**	$u = 1 + \sqrt{2x}$, $dx = (u-1)du$: $\int(1 - \frac{1}{u})du = \sqrt{2x} - \ln(1 + \sqrt{2x}) + c$.
11	**(a)**	$\frac{1}{(1-x)^2} = \sum(n+1)x^n$, $\frac{x}{(1-x)^2} = \sum nx^n$: total $\sum(2n+1)x^n$, $\lvert x\rvert < 1$.
12	**(a)**	Geometric, $a = 1$, $r = 0.3$: $\frac{1}{1-0.3} = \frac{10}{7}$.
13	**(c)**	$\sqrt[n]{n/3^n} = n^{1/n}/3 \to \frac{1}{3}$.
14	**(e)**	By parts with $v = -\frac{1}{2(2x+1)}$: result $\frac{e^{2x}}{4(2x+1)} + c$.

Q	Ans	Brief explanation
15	**(b)**	$L = \sqrt{2+L} \Rightarrow L^2 - L - 2 = 0$, and $L > 0$ gives $L = 2$.
16	**(a)**	$4x - x^2 = 4 - (x-2)^2$, so $2\arcsin\frac{x-2}{2} + c$.
17	**(b)**	Shells: $2\pi\int_0^2 (4-x)(2x-x^2)dx = 2\pi \cdot 4 = 8\pi$.
18	**(b)**	$u = x^2$: $\frac{1}{2}\int_0^4 \frac{e^u}{e^u+1}du = \frac{1}{2}\ln\frac{1+e^4}{2}$.
19	**(e)**	$u = \cosh(x-1)$: $\int u^2 du = \frac{1}{3}\cosh^3(x-1) + c$.
20	**(e)**	$\lvert S - S_N\rvert \le \frac{1}{(N+1)^3} < 0.001 \Rightarrow N+1 > 10$, so $N = 10$ terms.
21	**(d)**	$R = 9$, centre 4; $x = 13$ gives alternating harmonic (converges), $x = -5$ harmonic (diverges): $(-5, 13]$.
22	**(b)**	$\arctan x^2 = \sum \frac{(-1)^n x^{4n+2}}{2n+1}$; integrating on $[0, \frac{1}{2}]$ gives the stated series.
23	**(b)**	$\sqrt{1+\tan^2 x} = \sec x$: $L = [\ln(\sec x + \tan x)]_0^{\pi/3} = \ln(2+\sqrt{3})$.
24	**(b)**	$\frac{\ln(1+1/n)}{n^{-1/2}}$ by L'Hopital gives $\frac{2\sqrt{n}}{n+1} \to 0$.
25	**(b)**	$F'(x) = \sqrt{\sin x}\cos x$; at $\frac{\pi}{6}$: $\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{6}}{4}$.
26	**(b)**	Subtracting known terms: $\frac{Cx+D}{x^2+4} = \frac{2x+4}{x^2+4}$, so $C + D = 6$.
27	**(e)**	$\frac{1}{(3k-1)(3k+2)} = \frac{1}{3}\left(\frac{1}{3k-1} - \frac{1}{3k+2}\right)$ telescopes to $\frac{1}{6}$.
28	**(a)**	$u = 4 + x^2$: $\frac{1}{4}\int u^{1/2} du = \frac{1}{6}(4+x^2)^{3/2} + c$.